**Project Design Phase-II**
**Technology Stack (Architecture & Stack)**

| | |
|---|---|
| Date | 29-06-2026 |
| Team ID | 23A51A05M8 |
| Project Name | Metro Ticket Generating System in ServiceNow |
| Maximum Marks | 4 Marks |

**Technical Architecture:**

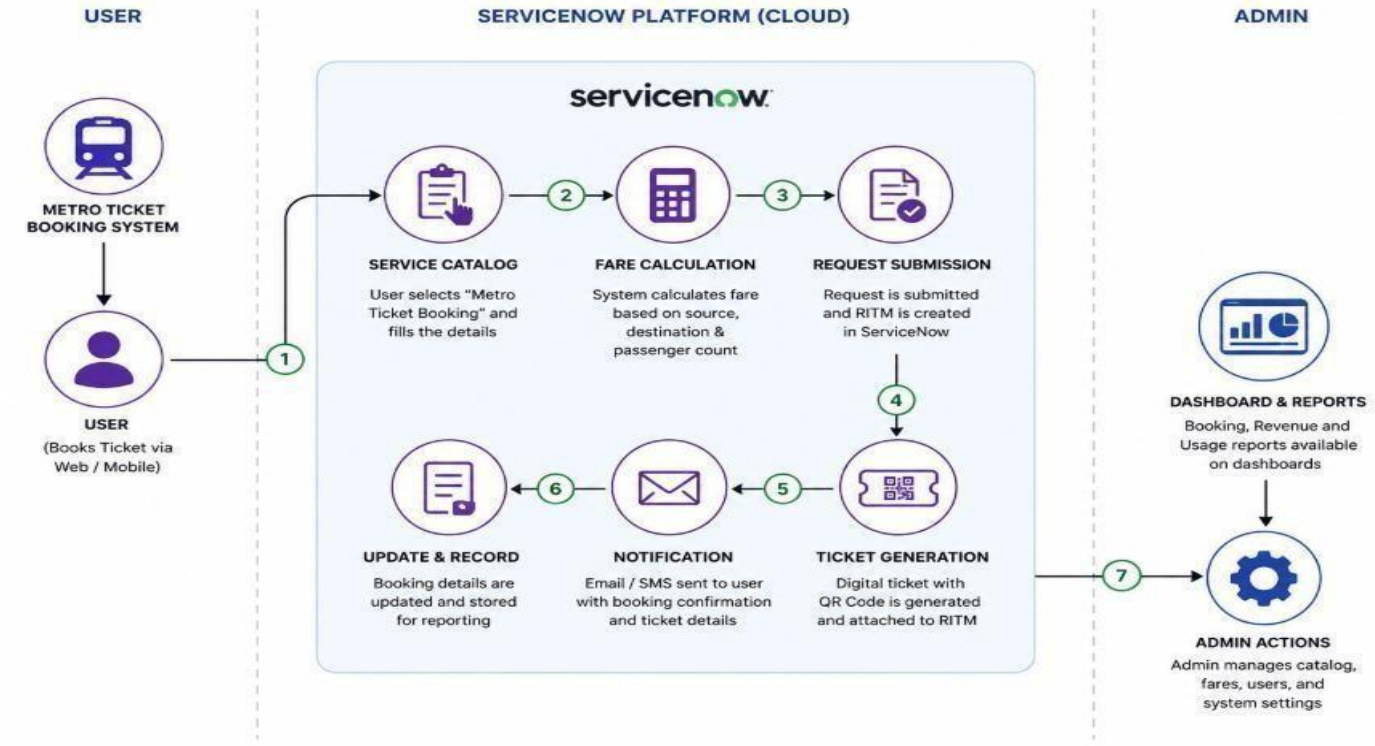


**Table-1 : Components & Technologies:**

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1. | User Interface | Interface for passengers to book metro tickets | ServiceNow Service Catalog, Portal UI |
| 2. | Application Logic-1 | Fare calculation based on source and destination stations | Catalog Client Scripts (JavaScript) |
| 3. | Application Logic-2 | Server-side fare retrieval and validation | Script Includes, GlideAjax |
| 4. | Application Logic-3 | Automated ticket booking workflow and notifications | Flow Designer, Business Rules |
| 5. | Database | Stores metro stations, fares, and booking information | ServiceNow Tables (u_metro_station, u_metro_fare, sc_req_item) |
| 6. | Cloud Database | Cloud-based data storage managed by ServiceNow | ServiceNow Cloud Database |
| 7. | File Storage | Stores QR codes, ticket attachments, and records | ServiceNow Attachments |
| 8. | External API-1 | QR Code generation for digital tickets | QR Code Generator API / Scripted QR Generation |
| 9. | External API-2 | Email notification service for ticket confirmation | ServiceNow Notification Engine |
| 10. | Machine Learning Model | Ticket booking trends, revenue, and usage reports | ServiceNow Reports & Dashboards |
| 11. | Infrastructure (Server / Cloud) | Application hosting and execution environment | ServiceNow SaaS Cloud Platform |

**Table-2: Application Characteristics:**

| S.No | Characteristics | Description | Technology |
|---|---|---|---|
| 1. | Open-Source Frameworks | Client-side scripting and customization used for fare calculation and form validation | JavaScript, GlideAjax |
| 2. | Security Implementations | Role-based access control for administrators and users, authenticated access through ServiceNow platform, secure data storage and transmission | ServiceNow ACLs, User Roles, HTTPS, IAM Controls |
| 3. | Scalable Architecture | ServiceNow's cloud-based multi-tenant architecture supports growing users and ticket bookings without infrastructure changes | ServiceNow SaaS Architecture, MultiTenant Cloud Platform |
| 4. | Availability | High availability through ServiceNow cloud infrastru managed services | ServiceNow Cloud Infrastructure, cture with continuous monitoring and Redundant Data Centers |
| 5. | Performance | Fast ticket booking process through server-side pro cloud-hosted services | GlideAjax, Script Includes, ServiceNow cessing, optimized database queries, and Database Optimization |

**References:** https://www.servicenow.com/docs/r/servicenow-platform/service-catalog/service-catalog.html https://www.servicenow.com/docs/r/application-development/building-applications/flow-designer.html https://www.servicenow.com/docs/r/api-reference/c_GlideAjaxAPI.html https://developer.servicenow.com https://www.servicenow.com/community/